

DIAGNOSTICS AND PLOTS FOR IDENTIFYING COLLINEARITY-INFLUENTIAL OBSERVATIONS AND FOR CLASSIFYING MULTIVARIATE OUTLIERS

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Summary: Simple, but very effective measures and plots are proposed to detect collinearity-influential observations in multiple linear regression analysis. Unlike traditional methods that are based mostly on the eigenstructure of the regressor matrix, these methods are based solely on the general tools of regression diagnostics, namely residuals and leverage values. They lead to a better understanding of influential observations and their effect on multivariate dispersion.

1. Introduction

Collinearities in multiple linear regression analysis are approximate linear dependencies among the column regressor values (Silvey, 1969 and Gunst, 1983). The major sources of collinearities are model constraints, restrictions on the values of the regressors due to the nature of the population, sampling deficiencies, overdefined models and outliers in the regressor space.

Denote by $\lambda(t, Z)$ (for the fixed t) a random variable, defined by the right-hand side in (14), where Z is a random variable with the following p.d.f.:

$$f_t(z) = \left\{ \lambda(t-z) \frac{\bar{F}((t-z)+z)}{\bar{F}(t-z)} \right\}, \quad z \in [0, t] \quad (15)$$

and an atom $(\bar{F}(t))$ at $z = t$. In accordance with definition (11), the following important expression for the mean conditional hazard rate is valid:

$$\begin{aligned} \lambda_{m,c}(t) &= \lambda(t) \int_0^t \theta(z) f_t(z) dz \\ &= \theta(t) f(t) + \int_0^t \lambda(t-z) \frac{\theta(z)}{\bar{F}(t-z)} dz, \end{aligned} \quad (16)$$

where an obvious relation: $\lambda(t) = f(t) / \bar{F}(t)$ was used.

The time in relations (14) and (16) is fixed and Z a random variable. For $t \geq 0$ the corresponding random process $\{Z(t), t \geq 0\}$ can be considered which in its turn defines the **hazard rate process** $\{\lambda(t, Z(t)), t \geq 0\} \equiv \{\lambda_t, t \geq 0\}$. The function $\lambda_{m,c}(t)$, $t \geq 0$ defines the mean of this process. It is well known (see Kebir, 1991, for instance) that for the hazard rate process the following equation for the survival function takes place:

$$P_f(t) = E \left[\exp \left\{ - \int_0^t \lambda_u du \right\} \right], \quad (17)$$

where a realization of λ_u in $[0, t]$ can be constructed in the following way.

Let the history for the Poisson process in $[0, t]$ prior to the point $u \in [0, t]$ be denoted by H_u and defined by the fixed ordered set of points: $H_t = \{t_1 < t_2 < \dots < t_k < t\}$. Then, conditioning on survival in the points prior

components of multivariate dispersion, and how they can be used to identify collinearity-influential observations.

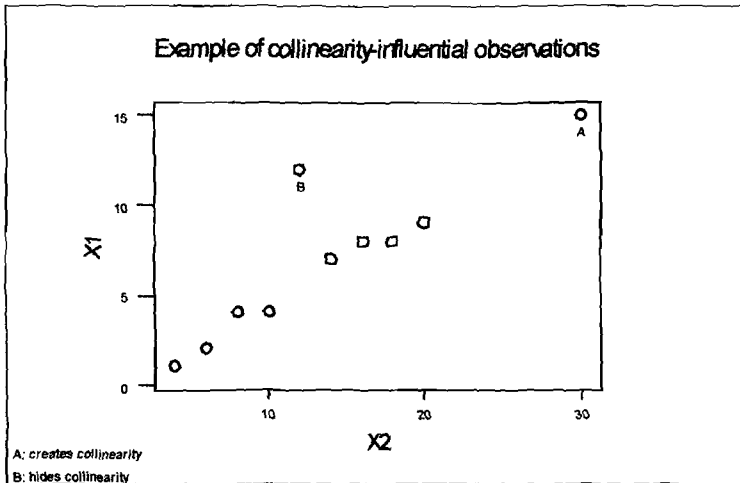


FIGURE 1

2. Detection of individual collinearity-influential observations

2.1 Introduction and Notation

Consider the regression model:

$$Y = X\beta + \epsilon \quad (1)$$

where Y is an $n \times 1$ vector of values of the dependent (response) variate, X is an $n \times p$ full rank matrix of regressors, including a constant term, β is a $p \times 1$ vector of unknown parameters (coefficients) and ϵ is an $n \times 1$ vector of independent random errors, each with zero mean and variance σ^2 . The hat

matrix is defined as

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \quad (2)$$

with diagonal elements

$$h_i = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i. \quad (3)$$

The diagonal elements (hat-diagonals) h_i are often used to identify leverage values. Similarly a measure of partial leverage (Chatterjee and Hadi, 1988) is defined as

$$\frac{e_{ix_j}^2}{\text{RSS}_{X_j}} = h_i - h_{i[j]} \quad (4)$$

$$= h_i - \mathbf{x}_{i[j]}'(\mathbf{X}_{[j]}'\mathbf{X}_{[j]})^{-1}\mathbf{x}_{i[j]}, \quad (5)$$

where the subscript $[j]$ indicates that the j th regressor has been omitted, while $e_{ix_j}^2$ and RSS_{X_j} are the i th squared residual and residual sum of squares respectively, which are obtained when the j th regressor $\mathbf{X}_j$ is regressed on the remaining regressors $\mathbf{X}_{[j]}$. The partial leverage represents the contribution of the j th regressor to the leverage of the i th observation. Let the j th centered regressor be denoted by

$$\bar{\mathbf{X}}_j = \mathbf{X}_j - \bar{\mathbf{X}}_j \mathbf{1}, \quad (6)$$

where $\bar{\mathbf{X}}_j = \bar{x}_j \mathbf{1}$.

The singular value decomposition (SVD) of $\mathbf{X}$ can be written as

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}', \quad (7)$$

where $\mathbf{D}$ is a $p \times p$ diagonal matrix with non-negative elements $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$, called the singular values of $\mathbf{X}$, $\mathbf{V}$ is a $p \times p$ matrix of p orthonormalized eigenvectors corresponding to the p eigenvalues of $\mathbf{X}'\mathbf{X}$, and $\mathbf{U}$ is an $n \times p$ matrix of p orthonormalized eigenvectors corresponding to the p non-zero eigenvalues of $\mathbf{X}\mathbf{X}'$. Based on the SVD of $\mathbf{X}$, the k th

condition index is defined as

$$\eta_k = \frac{\lambda_1}{\lambda_k}, \quad k = 2, \dots, p \quad (8)$$

of which the condition number $\kappa = \eta_p$ is a special case.

The following formal definition of collinearity-influential observations, in terms of the condition indices, was given by Walker (1989):

Definition. Let $\mathbf{x}_i$ denote the i th row of the $\mathbf{X}$ matrix and η_k and $\eta_{k(i)}$ the k th condition index computed with and without $\mathbf{x}_i$, respectively, then $\mathbf{x}_i$ is a collinearity-influential observation if, for a predetermined value δ ,

$$\frac{|\eta_k - \eta_{k(i)}|}{\eta_k} > \delta, \quad \text{for } k = 2, 3, \dots, p. \quad (9)$$

2.2 Diagnostics for individual collinearity-influential observations

2.2.1 A General Diagnostic

Hadi (1988) proposed a measure δ_i^* based on the condition number κ of $\mathbf{X}$:

$$\delta_i^* = \frac{\kappa_{(i)}^* - \kappa}{\kappa}, \quad (10)$$

where $\kappa_{(i)}^*$ is the condition number of $\mathbf{X}_{(i)}$, that is $\mathbf{X}$ without the i th case $\mathbf{x}_i$. Unfortunately the singular values of $\mathbf{X}_{(i)}$ cannot be obtained in a general closed form with respect to the singular values of $\mathbf{X}$. Because of the computational cost of the measure δ_i^* (the eigenvalues must be computed for $n+1$ matrices), Hadi proposed an approximation which leads to reasonable results. Note that a large positive value for δ_i^* points towards a type B outlier, because its deletion increases the condition index. Similarly a large negative

value for δ_i^* points towards a type A outlier, the deletion of which decreases the condition index. Belsley (1991) pointed out several weaknesses:

- δ_i^* is not properly scaled so that it is not most effective in measuring the degree of ill conditioning. He proposes that $\tilde{\delta}_i^*$, based on the scaled condition numbers $\tilde{\kappa}$ and $\tilde{\kappa}_{(i)}$, be used. The scaled condition numbers are based on singular values that result from applying the singular-value decomposition to $\mathbf{X}$ after its columns have first been scaled to have unit length. That is, if $\mathbf{X} = [\mathbf{X}_1 \dots \mathbf{X}_p]$, $s_i \equiv (\mathbf{X}_i' \mathbf{X}_i)^{-\frac{1}{2}}$, and $\mathbf{S} \equiv \text{diag}(s_1, \dots, s_p)$, then

$$\tilde{\kappa}(\mathbf{X}) = \kappa(\mathbf{XS})$$
- δ_i^* can be meaninglessly large for cases that are of no importance. Belsley uses the example of $\tilde{\kappa} = 6$ and $\tilde{\kappa}_{(i)} = 3$ which produces a large value of $\tilde{\delta}_i^*$, but is "most unexciting";
- δ_i^* is based only on the largest condition index, but ignores the rest; and
- δ_i^* is computationally expensive.

We add the following points to the list:

- δ_i^* does not distinguish between low-leverage (nonserious) observations and high-leverage (serious) type AB observations, that is, all noncollinearity-influential observations result in small values of δ_i^* , so that this information is lost;
- δ_i^* is confined to the interval $[-1; +\infty]$, which means that type A and type B observations are not treated equally;
- the plot of δ_i^* versus i is useful, but suffers from the same flaws as pointed out under the previous point;
- there are no cut-off values for type A and B observations, so that diagnosing serious cases remains somewhat subjective;
- certain collinearity-influential observations have partial influence in the sense that they usually do not affect all regressors. δ_i^* gives no indication of which regressors are involved in the collinearity, so that this information is lost;
- δ_i^* gives no indication of how multivariate dispersion, that is the scale (variance) and the orientation (correlation) of the regressors, is affected;
- δ_i^* cannot analytically be related to general measures of influence, so that

the relationship between influential observations and the conditioning of the regressor matrix is not clear.

A positive point about δ_i^* is that it can be generalized to detect multiple collinearity-influential observations. The diagnosis of multiple cases will have the same limitations as that of the single-observation method. Influential subsets are identified by the brute-force method of all possible subsets.

2.2.2 Leverage Components

Walker (1989) notes that $|\delta_i^*|$ can be related to the hat diagonals h_i , commenting: *"It seems, therefore, that a necessary condition for an observation to be collinearity-influential is to be a high leverage observation."*

She proposed the p terms of $h_i = \sum_{j=1}^p u_{ij}^2$ as collinearity-influential diagnostics, where $H = UU'$. The p terms are called "the leverage components" (LC) of $\mathbf{x}_i$ by Chatterjee and Hadi (1988), since they reflect the component of the leverage of $\mathbf{x}_i$ in the direction of the j th eigenvector.

The diagnostics are based on the assumption that a large u_{ik} suggests that the deletion of $\mathbf{x}_i$ will primarily affect the k th condition index. The cut-off for h_i is usually taken as $2p/n$, in the light of which Walker suggested considering a large u_{ik}^2 as one that accounts for at least half of this, that is when u_{ik}^2 is larger than p/n . Belsley (1991) points out that a "limitation to the value of this diagnostic" is that h_i cannot directly be related to the change in condition indices based on the scaled X matrix. We also note that most of the points of critique mentioned for δ_i^* also apply to the LC components. We focus on two only, namely:

- the classification of the various outliers as type A, B or AB is somewhat subjective, especially as far as type AB observations are concerned
- no indication is given of which regressors are involved in the collinearity

In the light of the previous exposition we set out to relate h_i not to the change in condition indices, but to the change in the variance inflation factors. This approach leads to a collinearity-influential measure that has few of the limitations of the measures discussed above.

3. An alternative approach to diagnosing collinearity-influential observations

3.1 The variance inflation factors

The variance inflation factors (Chatterjee and Price, 1977) are diagnostic tools often used to determine to what extent regressors are involved in collinearities.

They are defined as the diagonal elements of the inverse correlation matrix of the regressors, but can also be expressed as (Velleman and Welsch, 1981)

$$\begin{aligned} \text{VIF}_j &= \frac{1}{1 - R_{X_j}^2} \\ &= \frac{\sum_i \tilde{x}_{ij}^2}{\text{RSS}_{X_j}}, \end{aligned} \quad (11)$$

where $R_{X_j}^2$ is the coefficient of determination from the regression of X_j on the other regressors and $\tilde{x}_{ij}$ is the ij th element of $\tilde{X}_j$, while RSS_{X_j} is the residual sum of squares. (For regression through the origin, computation and interpretation of all these measures are different.) If X_j is orthogonal to the other regressors, $\text{VIF}_j = 1$, but a near-singularity involving X_j and the other regressors can result in a large VIF_j . Marquardt (1970) suggested a $\text{VIF}_j \geq 10$ as a guide to serious collinearity. The use of the term "variance inflation" is explained by the fact that the variance of the j th regression

coefficient is given by

$$\text{var}(\hat{\beta}_j) = \frac{\sigma^2}{\sum_i \tilde{x}_{ij}^2} \text{VIF}_j. \quad (12)$$

Although the VIF is often used to assess the effects of collinearity on the variance of estimators, it can also be a useful index of collinearity (Stewart, 1987). The mean of the VIF values (Chatterjee and Price, 1977) is another useful index of collinearity. If the $\mathbf{X}$ matrix is standardized so that $\mathbf{X}'\mathbf{X}$ is the correlation matrix of the regressors, then the total mean square error is

$$E \left[(\hat{\beta} - \beta)' (\hat{\beta} - \beta) \right] = \sigma^2 \sum_{j=1}^{p-1} \text{VIF}_j \quad (13)$$

If the regressors are orthogonal, then $\text{VIF}_j = 1$ and

$$E \left[(\hat{\beta} - \beta)' (\hat{\beta} - \beta) \right] = (p-1) \sigma^2 \quad (14)$$

The ratio

$$\frac{\sigma^2 \sum_{j=1}^{p-1} \text{VIF}_j}{\sigma^2 (p-1)} \quad (15)$$

measures the squared error in the least squares estimators relative to the size of the error for orthogonal data. This ratio is simply the mean of the VIF values

$$\overline{\text{VIF}} = \frac{\sum_{j=1}^{p-1} \text{VIF}_j}{p-1} \quad (16)$$

A value of $\overline{\text{VIF}}$ substantially larger than 1 is indicative of a serious collinearity problem. Marquardt and Snee (1975) regard the maximum VIF, say VIF_p , as "the best single measure of the conditioning of the data". This notion is supported by Berk (1977), who shows that

$$\text{VIF}_p \leq \kappa^2 < p (\text{VIF}_1 + \dots + \text{VIF}_p), \quad (17)$$

where κ^2 is the squared condition number of the matrix.

3.1.1 Centering and scaling

The issue of whether the regressors should be centered or not prior to determining collinearity structures has been debated for some time (Belsley, 1984). To avoid entering this dispute we adopt the middle of the road approach by calculating the VIF using the centered data if the model includes a constant term; otherwise the VIF is based on the uncentered data. This is also the approach used by SAS®.

The VIF, like the condition number, is invariant under scaling.

3.2 A decomposition of the VIF

The size of the VIF_j being accepted as a measure of collinearity, we are now concerned with how it may be affected by individual observations. If $VIF_{(i)j}$ denotes the j th VIF after deleting the i th observation, then $VIF_{(i)j}$ can be expressed as follows:

$$\begin{aligned}
 VIF_{(i)j} &= \frac{\tilde{\mathbf{x}}'_{(i)j} \tilde{\mathbf{x}}_{(i)j}}{RSS_{X_{(i)j}}} \\
 &= \frac{\sum_i \tilde{x}_{ij}^2 - \left(\frac{n}{n-1}\right) \tilde{x}_{ij}^2}{RSS_{X_j} - e_{iX_j}^2 / (1 - h_{i[j]})} \\
 &= \frac{\sum_i \tilde{x}_{ij}^2 / RSS_{X_j} - \left(\frac{n}{n-1}\right) \tilde{x}_{ij}^2 / RSS_{X_j}}{1 - e_{iX_j}^2 / RSS_{X_j} (1 - h_{i[j]})} \\
 &= \frac{VIF_j - \left(\frac{n}{n-1}\right) \tilde{x}_{ij}^2 / RSS_{X_j}}{(1 - h_i) / (1 - h_{i[j]})}
 \end{aligned}$$

$$\begin{aligned}
&= \frac{\text{VIF}_j \left(1 - \left(\frac{n}{n-1} \right) \frac{\tilde{x}_{ij}^2}{\sum_i^n \tilde{x}_{ij}^2} \right)}{(1 - h_i) / (1 - h_{i[j]})} \\
&= \text{VIF}_j \left(\frac{1 - h_{i[j]}}{1 - h_i} \right) \left(1 - \left(\frac{n}{n-1} \right) \frac{\tilde{x}_{ij}^2}{\sum_i^n \tilde{x}_{ij}^2} \right) \\
&= \text{VIF}_j O_{i_{X_j}} S_{i_{X_j}}^{-1}, \tag{18}
\end{aligned}$$

where

$$O_{i_{X_j}} = \left(\frac{1 - h_{i[j]}}{1 - h_i} \right) \tag{19}$$

and

$$S_{i_{X_j}} = \left(1 - \left(\frac{n}{n-1} \right) \frac{\tilde{x}_{ij}^2}{\sum_i^n \tilde{x}_{ij}^2} \right)^{-1} \tag{20}$$

The factor $O_{i_{X_j}}$ can be viewed as the capacity or potential of $\mathbf{x}_i$ to *orientate* $\mathbf{X}_j$ away from a collinearity. Note that $O_{i_{X_j}}$ is monotone increasing in h_i for a fixed value of $h_{i[j]}$. This implies that type B observations can have large values of $O_{i_{X_j}}$ relative to $S_{i_{X_j}}$, as h_i tends to one and $h_{i[j]}$ tends to zero. The factor $S_{i_{X_j}}$ can be viewed as the (raw) *scale inflation* of $\mathbf{X}_j$ due to the i th observation. $S_{i_{X_j}}$ can be large relative to $O_{i_{X_j}}$ in the case of type A observations.

Since both $O_{i_{X_j}} \geq 1$ and $S_{i_{X_j}} \geq 1$, it is clear that $\text{VIF}_{(i)j} \geq \text{VIF}_j$ if and only if $O_{i_{X_j}} \geq S_{i_{X_j}}$. This is due to type B observations which deflate variances and covariances, and thus also the VIF. A straightforward deduction is that for type B observations $O_{i_{X_j}} > S_{i_{X_j}}$ and for type A observations $S_{i_{X_j}} > O_{i_{X_j}}$.

3.3 A proposed diagnostic

We propose the measure $\varphi_{i_{x_j}}$ to detect collinearity-influential observations and to classify high leverage points, where

$$\begin{aligned}\varphi_{i_{x_j}} &= \frac{\text{VIF}_{(i)j} - \text{VIF}_j}{\max(\text{VIF}_j; \text{VIF}_{(i)j})} \\ &= \frac{\frac{O_{i_{x_j}}}{S_{i_{x_j}}} - 1}{\max\left(\frac{O_{i_{x_j}}}{S_{i_{x_j}}}; 1\right)}\end{aligned}\quad (21)$$

which reduces to

$$\varphi_{i_{x_j}} = 1 - \frac{S_{i_{x_j}}}{O_{i_{x_j}}} \quad (22)$$

for type B observations and

$$\varphi_{i_{x_j}} = \frac{O_{i_{x_j}}}{S_{i_{x_j}}} - 1 \quad (23)$$

for type A observations.

$\varphi_{i_{x_j}}$ when used in conjunction with h_i , has several attractive properties as a diagnostic of collinearity-influential observations:

- Observations of interest are high leverage observations with relatively large values of $\left|\varphi_{i_{x_j}}\right|$.
- For high leverage observations that are not collinearity-influential (i.e. serious type AB observations), $\varphi_{i_{x_j}}$ will take on values close to zero as $O_{i_{x_j}} \approx S_{i_{x_j}}$.
- Low leverage observations will have small values of h_i and hence will have values of $\varphi_{i_{x_j}} \approx 0$.
- It can be shown that $\left|\varphi_{i_{x_j}}\right| \leq h_i$ (see Appendix 1) so that a high leverage observation (large h_i) is also a (serious) collinearity-influential observation if $\left|\varphi_{i_{x_j}}\right| \rightarrow h_i$.
- A clear distinction is made between type A and type B observations as

φ_{iX_j} takes on negative values for type A and positive values for type B observations.

- φ_{iX_j} is bounded by 1 in the case of type B observations, and by -1 in the case of type A observations. Hence both type A and type B observations are measured on an absolute scale from 0 to 1 as $h_i \rightarrow 1$. In general tighter bounds for φ_{iX_j} are given by $[-h_i; h_i]$. To exclude trivial cases, we propose the cutoff $2p/n$ for $|\varphi_{iX_j}|$.
- Regressors of which the correlation structures are affected by the i th collinearity-influential observation are clearly identified by having large (absolute) values of $(h_i; |\varphi_{iX_j}|)$.
- A simple, but very effective graphical tool is a plot of φ_{iX_j} versus h_i with reference lines for φ_{iX_j} at $+45^\circ$ (for type B observations), 0 (for type AB observations) and -45° (for type A observations). This plot resembles a megaphone (see Figures 2 and 3). Serious type A, type B and type AB observations will be separated from the bulk of the observations, having large values of h_i . Type AB observations will be close to or on the 0 reference line, while the seriousness of a collinearity-influential observation can be ascertained by the closeness of $|\varphi_{iX_j}|$ to h_i .
- φ_{iX_j} is computationally inexpensive. It is a function of S_{iX_j} , which is based on univariate descriptive statistics for the j th regressor, and the factor O_{iX_j} . The latter seems to pose a computational problem, requiring the computation of the partial leverage $(h_i - h_{i[j]})$. This is, however, not the case, since the partial leverage can be computed from the elements of the single regression fit, as shown by Velleman and Welsch (1981).
- φ_{iX_j} is based on the VIF, which is an accepted measure of collinearity.
- φ_{iX_j} reflects on aspects of scale and orientation.
- The values h_i and $h_{i[j]}$ are basic tools for most measures of partial influence, that is the influence of the i th observation on the j th regressor. Well-known examples of measures of the influence of an observation on a single coefficient are D_{ij} (Cook and Weisberg, 1980), where

$$D_{ij} = \frac{e_i^2}{\hat{\sigma}^2 (1 - h_i)} \frac{(h_i - h_{i[j]})}{(1 - h_i)}$$

$$\begin{aligned}
 &= \frac{e_i^2}{\hat{\sigma}^2 (1 - h_i)} (O_{ix_j} - 1) \\
 &= r_{iY}^2 (O_{ix_j} - 1)
 \end{aligned} \tag{24}$$

and DFBETAS_{ij} (Belsley, Kuh and Welsch, 1980), where

$$\begin{aligned}
 \text{DFBETAS}_{ij}^2 &= \frac{e_i^2}{\hat{\sigma}_{(i)}^2 (1 - h_i)} \frac{(h_i - h_{i[j]})}{(1 - h_i)} \\
 &\quad \frac{e_i^2}{\hat{\sigma}_{(i)}^2 (1 - h_i)} (O_{ix_j} - 1) \\
 &= t_{iY}^2 (O_{ix_j} - 1).
 \end{aligned} \tag{25}$$

These examples illustrate the dependence of measures of partial influence on the factor O_{ix_j} . Type B and to a lesser extent type AB observations hence can also be (partial) influential observations which emphasize that partial-influential observations can be collinearity influential, but need not be.

- Finally φ_{ix_j} can be generalized to φ_{IX_j} to detect multiple collinearity-influential observations. Influential subsets are identified by the brute-force method of all possible subsets or by deleting similar observations (type A or B) identified by φ_{ix_j} .
- Let the observations that are to be deleted be indexed by i . The VIF can readily be factorized to accommodate the deletion of multiple observations as follows:

$$\begin{aligned}
 \text{VIF}_{(i)X_j} &= \frac{\tilde{\mathbf{X}}_j' \tilde{\mathbf{X}}_j - \tilde{\mathbf{X}}_{ij}' (\mathbf{I} - \mathbf{V}_i)^{-1} \tilde{\mathbf{X}}_{ij}}{\text{RSS}_{X_j} - \mathbf{e}_{iX_j}' (\mathbf{I} - \mathbf{H}_{i[j]})^{-1} \mathbf{e}_{iX_j}} \\
 &= \text{VIF}_{X_j} \left[1 - \frac{\mathbf{e}_{iX_j}' (\mathbf{I} - \mathbf{H}_{i[j]}) \mathbf{e}_{iX_j}}{\text{RSS}_{X_j}} \right]^{-1} \left[1 - \frac{\tilde{\mathbf{X}}_{ij}' (\mathbf{I} - \mathbf{V}_i)^{-1} \tilde{\mathbf{X}}_{ij}}{\tilde{\mathbf{X}}_j' \tilde{\mathbf{X}}_j} \right] \\
 &= \text{VIF}_{X_j} O_{iX_j} S_{iX_j}^{-1}
 \end{aligned} \tag{26}$$

where

$$\mathbf{H}_{i[j]} = \mathbf{X}_{i[j]} \left(\mathbf{X}'_{[j]} \mathbf{X}_{[j]} \right)^{-1} \mathbf{X}'_{i[j]} \quad (27)$$

and $(\mathbf{I} - \mathbf{V}_1)$ is the submatrix formed by the intersection of the rows and columns indexed by i of the matrix (Cook and Weisberg, 1982), that is

$$\mathbf{I} - \mathbf{V} = \begin{pmatrix} 1 - \frac{1}{n} & -\frac{1}{n} & \dots & -\frac{1}{n} \\ -\frac{1}{n} & 1 - \frac{1}{n} & \dots & -\frac{1}{n} \\ \dots & \dots & \dots & \dots \\ -\frac{1}{n} & -\frac{1}{n} & \dots & 1 - \frac{1}{n} \end{pmatrix} \quad (28)$$

Note that this factorization is a generalization of (18).

- φ_{ix_j} is scale invariant.

4. Examples of collinearity-influential observations

4.1 Example 1

The first data set is from Hocking and Pendleton (1983). This artificial data set consists of 26 observations on 3 regressors and a response. The fitted model includes a constant term. The regressor X_3 was created as a linear function of X_1 and X_2 as follows:

$$2X_3 = 60 - 3X_1 - 1.5X_2 + e_2, \quad (29)$$

where $e_2 \sim N(0; 0.16)$. The collinearity between X_1 , X_2 and X_3 is obscured by a type B observation number 24, which has high leverage $h_i = 0.927$. The VIFs for X_1 , X_2 and X_3 are 1.769, 1.86 and 2.82 respectively.

If observation 24 is omitted, the collinearity is revealed, as the VIFs then become 14.292, 14.563 and 30.637.

The value of the measure δ_i^* is

$$\begin{aligned}\delta_i^* &= \frac{\kappa_{(i)}^* - \kappa}{\kappa} \\ &= \frac{132.45622 - 43.32288}{43.32288} \\ &= 2.05742.\end{aligned}$$

The leverage components (LC) for observation 24 are given by table 1.

Table 1. Leverage components for the Hocking-Pendleton data

	u_1^2	u_2^2	u_3^2	u_4^2
LC	0.057	0.015	0.167	0.687

Now $\delta_i^* = 2.05742$ points towards a type B observation, but gives no indication whether it is serious. The leverage component $u_4^2 = 0.687$ exceeds the proposed size-adjusted cutoff $p/n = 0.1538$, thus pointing to a type B observation. The value $u_4^2 = 0.687$ is, however, not close to the leverage value $h_{24} = 0.927$, which makes it difficult to decide whether this is a serious case. Furthermore there is no indication of which regressors are affected.

Table 2 summarizes the measures φ_{ix_j} , O_{ix_j} and S_{ix_j} introduced in this paper.

Table 2. Measures related to the VIF for the Hocking-Pendleton data

	X_1	X_2	X_3
h_{24}	0.927	0.927	0.927
φ_{24x_j}	0.876	0.872	0.908
O_{24x_j}	8.610	8.171	11.975
S_{24x_j}	1.066	1.044	1.102

Observation 24 is exposed as being a type B collinearity-influential observation, obscuring a collinearity between X_1 , X_2 and X_3 . All three of the φ_{24X_j} exceed the cutoff $2p/n = 0.3077$. Note that X_3 is pointed out as being the most seriously affected, having the largest value $\varphi_{24X_3} = 0.908$. This is indeed the case, since X_3 was generated as a linear combination of X_1 and X_2 . Observation 24 potentially has the capacity of being a serious collinearity-influential observation, because of the high leverage $h_{24} = 0.927$. This is confirmed by the large positive values of φ_{24X_j} , which exposes it as a type B observation that hides a collinearity amongst X_1 , X_2 and X_3 . Note that $\varphi_{24X_3} = 0.908$, which is close to $h_{24} = 0.927$. This emphasizes the fact that observation 24 is highly influential on the correlation structure of the regressors. Furthermore

$$\begin{aligned} \text{VIF}_{(24)X_3} &= \text{VIF}_{X_3} \left(\frac{O_{24X_3}}{S_{24X_3}} \right) \\ &= 2.82 \left(\frac{11.975}{1.102} \right) \\ &= 30.64. \end{aligned}$$

The relatively large value of $O_{24X_3} = 11.975$ versus $S_{24X_3} = 1.102$ suggests that observation 24 lies far off the fitted plane defined by X_1 and X_2 , whereas it is not extreme, relative to its regressor means.

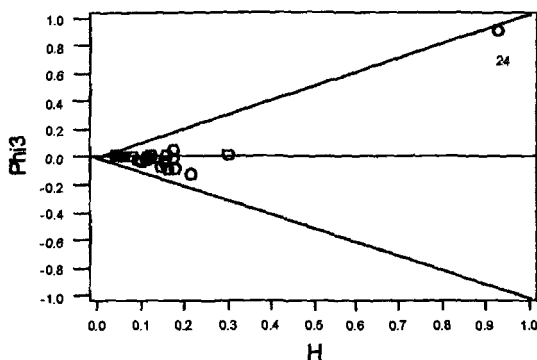


Figure 2. Plot of φ_{iX_3} versus h_i for the Hocking-Pendleton data

4.2 Example 2

The second data set was compiled by Gunst and Mason (1980). The 7 variables on 49 countries are

GNP	(Y)	– Gross National Product per Capita, 1957, U.S. Dollars
INFD	(X ₁)	– Infant Deaths per 1,000 Live Births
PHYS	(X ₂)	– Number of Inhabitants per Physician
DENS	(X ₃)	– Population per square Kilometer
AGDS	(X ₄)	– Population per 1,000 Hectares of Agricultural Land
LIT	(X ₅)	– Percentage Literates of Population Aged 15 and Over
HIED	(X ₆)	– Students Enrolled in Higher Education per 100,000 Population

An analysis of the data reveals a strong collinearity between X_3 and X_4 , and there are several high leverage observations. The VIFs for X_3 and X_4 are 19.098 and 18.847 respectively.

The values for δ_i^* and leverage components (LC) are given by Table 3.

Table 3. Leverage components and values for δ_i^* for the Gunst-Mason Data

Country	h_i	u_1^2	u_2^2	u_3^2	u_4^2	u_5^2	u_6^2	u_7^2	δ_i^*
HongKong	0.511	0.441	0.000	0.006	0.037	0.003	0.000	0.023	0.011
India	0.558	0.000	0.099	0.008	0.001	0.000	0.435	0.016	0.012
Malta	0.688	0.003	0.002	0.000	0.612	0.013	0.002	0.057	0.139
Singapore	0.632	0.536	0.010	0.000	0.074	0.000	0.002	0.013	0.010
UnitedStates	0.490	0.000	0.003	0.319	0.009	0.147	0.010	0.001	-0.001

The values of δ_i^* , with perhaps the exception of Malta, are disappointing. They even take on small positive values for type A observations in the case of Hong Kong and Singapore. The leverage components perform better with all of the applicable u_i^2 exceeding the cutoff $p/n = 0.1429$. However, it seems that Malta, a type B observation, can be classified as type AB, whereas India, a type AB observation, can be classified as B.

Table 4 contains a summary of the measures $\varphi_{i_{X_j}}$, $O_{i_{X_j}}$ and $S_{i_{X_j}}$.

Table 4. Measures based on the VIF for the Gunst-Mason Data

Country		X_1	X_2	X_3	X_4	X_5	X_6
Hong Kong	h_{17}	0.511	0.511	0.511	0.511	0.511	0.511
	φ_{17x_j}	0.004	0.011	-0.422	-0.429	0.016	-0.008
	O_{17x_j}	1.005	1.029	1.104	1.008	1.037	1.002
	S_{17x_j}	1.001	1.018	1.911	1.765	1.021	1.010
India	h_{20}	0.558	0.558	0.558	0.558	0.558	0.558
	φ_{20x_j}	-0.112	-0.086	0.002	0.000	-0.151	0.031
	O_{20x_j}	1.823	1.011	1.003	1.001	1.000	1.039
	S_{20x_j}	2.052	1.106	1.001	1.001	1.178	1.007
Malta	h_{27}	0.688	0.688	0.688	0.688	0.688	0.688
	φ_{27x_j}	0.037	0.114	0.609	0.625	0.157	-0.013
	O_{27x_j}	1.041	1.136	2.657	2.669	1.211	1.000
	S_{27x_j}	1.003	1.007	1.039	1.000	1.021	1.013
Singapore	h_{39}	0.632	0.632	0.632	0.632	0.632	0.632
	φ_{39x_j}	-0.001	0.031	-0.297	-0.298	-0.004	0.002
	O_{39x_j}	1.001	1.034	1.209	1.505	1.034	1.002
	S_{39x_j}	1.006	1.002	1.720	2.145	1.038	1.000
United States	h_{46}	0.490	0.490	0.490	0.490	0.490	0.490
	φ_{46x_j}	-0.001	-0.007	0.000	0.002	-0.002	0.047
	O_{46x_j}	1.000	1.003	1.003	1.004	1.012	1.883
	S_{46x_j}	1.010	1.010	1.003	1.002	1.014	1.799

The exposed collinearity-influential observations on regressors X_3 and X_4 are Hong Kong, Singapore and Malta. They have values of $\varphi_{i\mathbf{x}_s}$ and $\varphi_{i\mathbf{x}_4}$ that exceed the proposed cutoff $2p/n = 0.2857$. Individually these observations are not that extreme as can be deduced from the relatively

small values of φ_{iX_j} compared to h_i , especially in the case of Singapore. Furthermore none of the h_i approaches one.

It can be concluded that India and the United States are not collinearity-influential observations on regressors X_3 and X_4 . While they are leverage values, they have values for φ_{iX_3} and φ_{iX_4} that tend to zero. This is confirmed by Figure 4.

Note that

$$\begin{aligned} \text{VIF}_{(27)X_3} &= \text{VIF}_{X_3} \left(\frac{O_{27X_3}}{S_{27X_3}} \right) \\ &= 19.098 \left(\frac{2.657}{1.039} \right) \\ &= 48.839, \end{aligned}$$

while $\text{VIF}_{(27)X_4} = 50.303$.

Similarly

$$\begin{aligned} \text{VIF}_{(17)X_3} &= \text{VIF}_{X_3} \left(\frac{O_{17X_3}}{S_{17X_3}} \right) \\ &= 19.098 \left(\frac{1.104}{1.911} \right) \\ &= 11.033, \end{aligned}$$

while $\text{VIF}_{(17)X_4} = 10.763$, $\text{VIF}_{(39)X_3} = 13.424$, and $\text{VIF}_{(39)X_4} = 13.224$.

The omission of observations 17 and 39 individually does not seem to remove

the collinearity between X_3 and X_4 completely. This is because of the masking effect the one has on the other (Mason and Gunst, 1985).

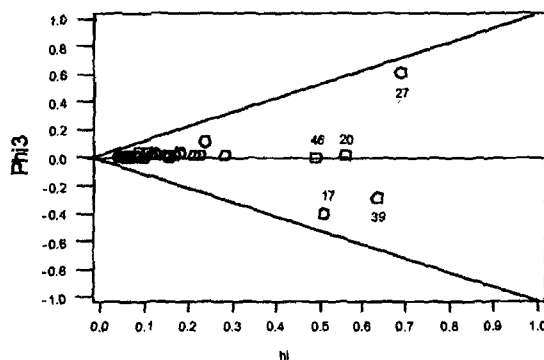


Figure 3. Plot of φ_{iX_3} versus h_i for the Gunst-Mason Data

4.3 An example of multiple collinearity-influential observations

Consider the Gunst and Mason data set. The simultaneous deletion of observations 17 and 39 leads to the results summarized in Table 5.

Table 5. φ_{IX_j} , O_{IX_j} and S_{IX_j} for the Gunst-Mason Data

	X_3	X_4
φ_{IX_j}	-0.859	-0.858
O_{IX_j}	1.641	11.610
S_{IX_j}	11.655	81.736

If $i = (17, 39)'$, then

$$\begin{aligned} \text{VIF}_{(i)X_3} &= \text{VIF}_{X_3} \left(\frac{O_{iX_3}}{S_{iX_3}} \right) \\ &= 19.098 \left(\frac{1.641}{11.655} \right) \\ &= 2.689. \end{aligned}$$

Similarly $\text{VIF}_{(i)X_4} = 2.677$. Thus deleting both observations 17 and 39 removes the collinearity completely. Finally, this example illustrates how the proposed methods are also appropriate in the case of multiple collinearity-influential observations.

5. Appendix 1

Proof that $|\varphi_{iX_j}| \leq h_i$.

Proof.

Let the design matrix be given by

$$\mathbf{X} = [\mathbf{X}_0 \mathbf{X}_1 \dots \mathbf{X}_{p-1}]$$

where $\mathbf{X}_0$ is assumed to be the constant term. Then

$$h_i = h_{i0} + h_{i1|0} + \dots + h_{ip-1|0 \dots p-2}$$

with

$$h_{i0} = \frac{1}{n}$$

and

$$h_{i1|0} = \frac{(x_{i1} - \bar{x}_1)^2}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}.$$

We first establish the maximum and minimum values of $O_{i_{X_j}}$ and $S_{i_{X_j}}$.

Now

$$\begin{aligned} O_{i_{X_j}} &= \frac{(1 - h_{i[j]})}{(1 - h_i)} \\ &\leq \frac{(1 - h_{i0})}{(1 - h_i)} \\ &= \frac{(1 - 1/n)}{(1 - h_i)} \\ &= \left(\frac{n-1}{n} \right) \frac{1}{1 - h_i}, \end{aligned}$$

so that the maximum value of $O_{i_{X_j}}$ is $\left(\frac{n-1}{n} \right) \frac{1}{1 - h_i}$. The minimum value of $O_{i_{X_j}} = 1$ for $h_{i[j]} = h_i$.

Similarly

$$\begin{aligned} S_{i_{X_j}} &= \frac{(1 - h_{i0})}{1 - (h_{i0} + h_{ij|0})} \\ &\leq \frac{(1 - h_{i0})}{(1 - h_i)} \\ &= \frac{(1 - 1/n)}{(1 - h_i)} \\ &= \left(\frac{n-1}{n} \right) \frac{1}{1 - h_i}, \end{aligned}$$

so that the maximum value of $S_{i_{X_j}}$ is $\left(\frac{n-1}{n} \right) \frac{1}{1 - h_i}$. The minimum value of $S_{i_{X_j}} = 1$ for $h_{ij|0} = 0$.

Let $O_{i_{x_j}} \geq S_{i_{x_j}}$, then

$$\begin{aligned}
 \left| \varphi_{i_{x_j}} \right| &= \varphi_{i_{x_j}} \\
 &= 1 - \frac{S_{i_{x_j}}}{O_{i_{x_j}}}, \text{ for } O_{i_{x_j}} \geq S_{i_{x_j}} \\
 &\leq 1 - \frac{\min(S_{i_{x_j}})}{\max(O_{i_{x_j}})} \\
 &= 1 - \frac{1}{\max(O_{i_{x_j}})} \\
 &= 1 - \frac{1}{(n-1)/n(1-h_i)^{-1}} \\
 &= 1 - \left(\frac{n}{n-1} \right) (1-h_i) \\
 &\leq h_i
 \end{aligned}$$

The above result readily follows when $O_{i_{x_j}} \leq S_{i_{x_j}}$, because of symmetry.

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References

- BELSLEY, D. (1991). *Conditioning diagnostics: Collinearity and weak data in regression*. John Wiley and Sons.
- BELSLEY, D.A. (1984). Demeaning conditioning diagnostics through centering (with discussions). *The American Statistician*, **38**(2):73–93.
- BELSLEY, E., KUH, D.A. & WELSCH, R.E. (1980). *Regression diagnostics: Identifying influential data and sources of collinearity*. John Wiley and Sons, New York.
- BERK, K.N. (1977). Tolerance and conditioning in regression computations. *Journal of the American Statistical Association*, **72**(360):863–866.
- CHATTERJEE, S. & HADI, A.S. (1988). *Sensitivity analysis in linear regression*. John Wiley and Sons, New York.
- CHATTERJEE, S. & PRICE, B. (1977). *Regression analysis by example*. John Wiley and Sons, New York.
- COOK, R.D. & WEISBERG, S. (1980). Characterizations of an empirical influence function for detecting influential cases in regression. *Technometrics*, **22**(4):495–508.
- COOK, R.D. & WEISBERG, S. (1982). *Residuals and influence in regression*. Chapman and Hall, London.
- FREUND, R.J. & RAMON, C.L. (1986). *SAS System for regression*. Institute Inc., Cary, NC, USA.
- GNANADESIKAN & KETTENRING. (1972). Robust estimates, residuals and outlier detection with multiresponse data. *Biometrics*, **28**:81–124.
- GUNST, R.F. (1983). Regression analysis with multicollinear predictor variables: Definition, detection, and effects. *Communications in Statistics – Theory and Methods*, **12**(19):2217–2260.
- GUNST, R.F. & MASON, R.L. (1980). *Regression analysis and its application: A data-oriented approach*. Marcel Dekker, Inc., New York.
- HADI, A.S. (1988). Diagnosing collinearity-influential observations. *Computational Statistics and Data Analysis*, **7**:143–159.

- HADI, A.S. & VELLEMAN, P.F. (1987). Comment: Diagnosing near collinearities in least squares regression. *Statistical Science*, 2(1):93–98.
- HOCKING, R.R. & PENDLETON, O.J. (1983). The regression dilemma. *Communications in Statistics – Theory and Methods*, 12(5):497–527.
- MARQUARDT, D.W. (1970). Generalized inverses, ridge regression, biased linear estimation and nonlinear estimation. *Technometrics*, 12(3):591–612.
- MARQUARDT, D.W. & SNEE, R.D. (1975). Ridge regression in practice. *The American Statistician*, 29(1):3–20.
- MASON, R.L. & GUNST, R.F. (1985). Outlier-induced collinearities. *Technometrics*, 27(4):401–407.
- SENGUPTA, D. & BHIMASANKARAM, P. (1997). On the roles of observations in collinearity in the linear model. *Journal of the American Statistical Association*, 92(439):1024–1033.
- SILVEY, S.D. (1969). Multicollinearity and imprecise estimation. *Journal of the Royal Statistical Society (B)*, 31:539–552.
- STEWART, G.W. (1987). Collinearity and least squares regression. *Statistical Science*, 2(1):68–100.
- VELLEMAN, P.F. & WELSCH, R.E. (1981). Efficient computing of regression diagnostics. *The American Statistician*, 35(4):234–242.
- WALKER, E. (1989). Detection of collinearity-influential observations. *Communications in Statistics – Theory and Methods*, 18(5):1675–1690.